

Deep Learning for Image Analysis

Mehyar MLAWEH



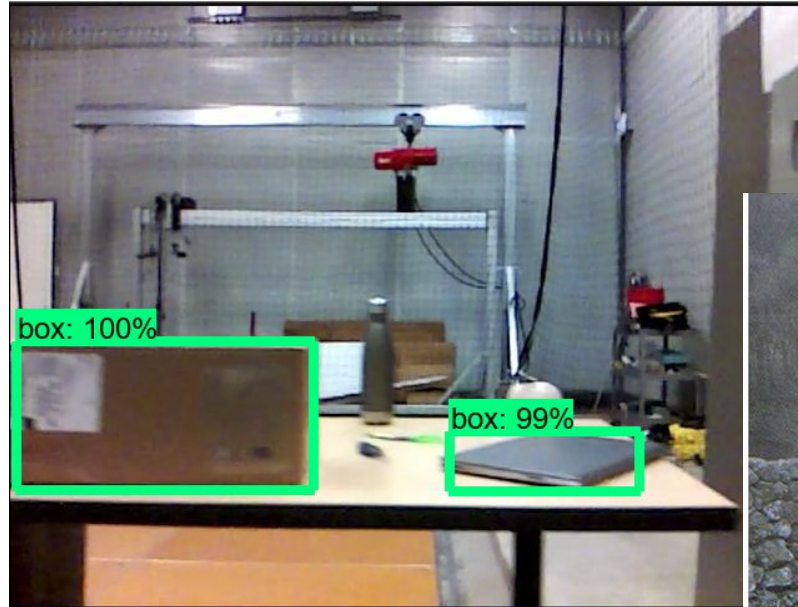
Master BDIA - Dauphine Tunis
PhD student - Université PSL

1. Computer Vision Foundations
2. CNNs
3. Transfer Learning
4. Vision Transformers
5. Object Detection
6. Segment Anything Model (SAM)
7. Final Project

Lecture 6 :

Segment Anything





Original image



Original image



Classification



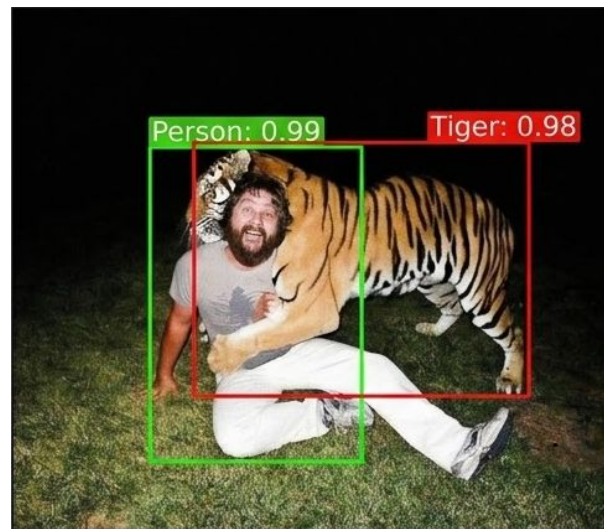
A person- 81%

No location. No boundary

Original image



Object Detection

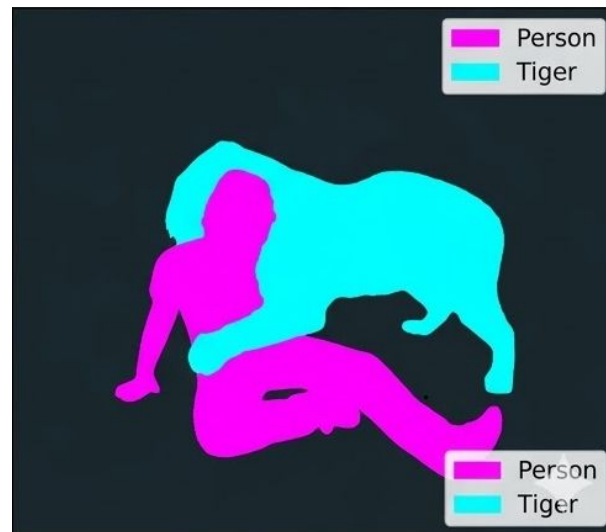


Bounding boxes around the
tiger and then person
Approximate location

Original image



Segmentation

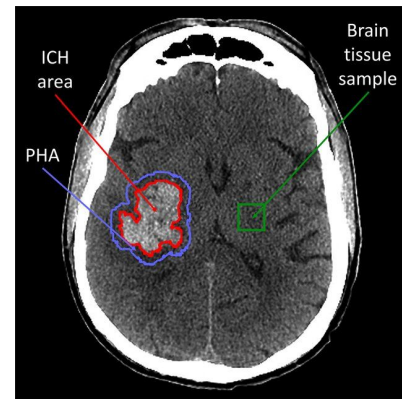


Precise pixel masks

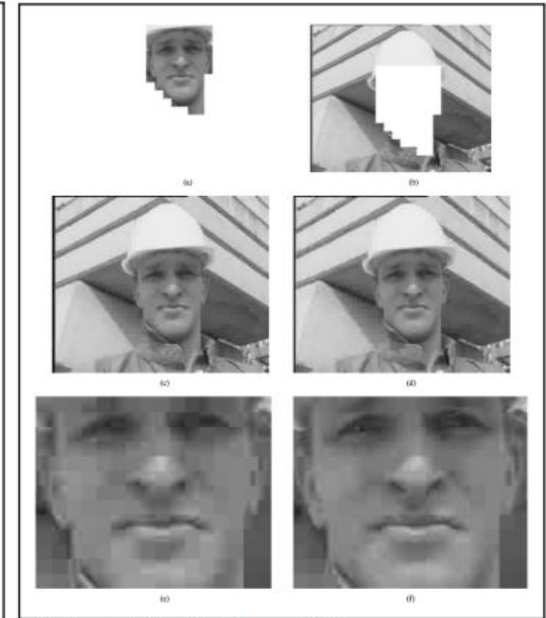
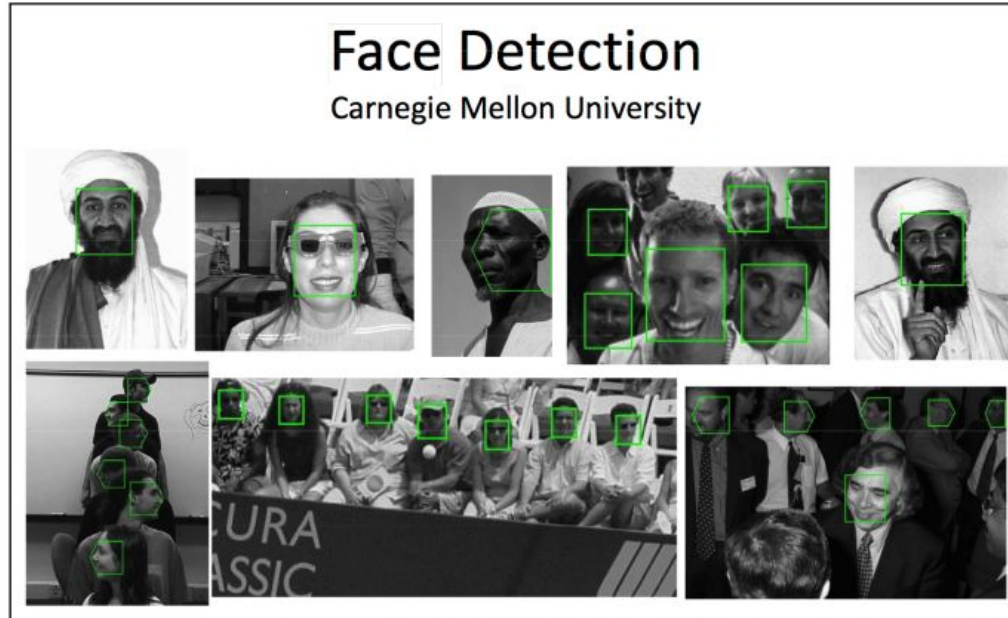
Assigning a label to every single pixel in the image

Assigning a label to every single pixel in the image

Not where **roughly**, where **exactly**

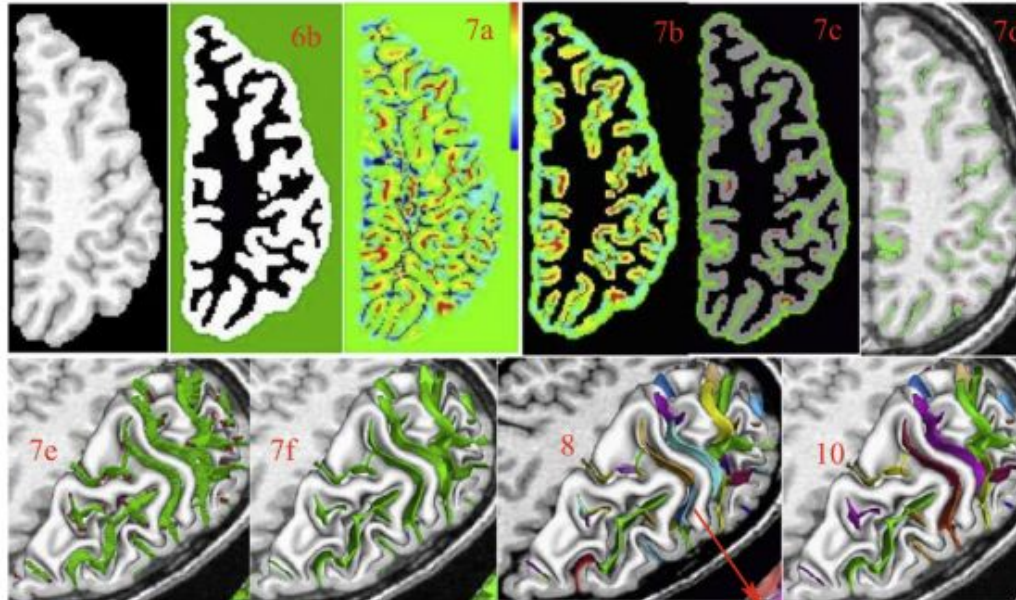


Some examples



Face Segmentation Using Skin-Color Map in Videophone Applications, Douglas Chai, and King N. Ngan, 1999

Some examples



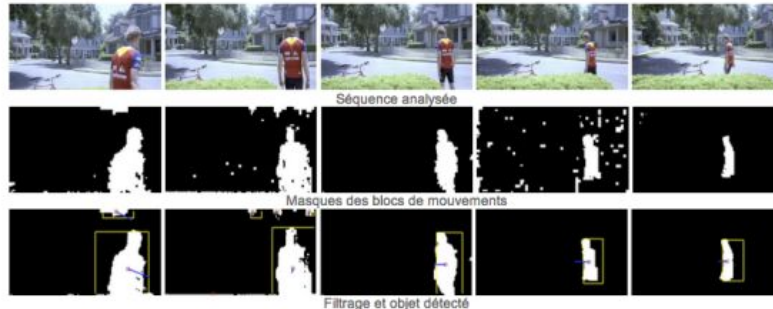
- Quantification of tissue and organ volumes
- Localization of a pathology
- Treatment planning
- ..

A framework to study the cortical folding patterns, Mangin et al., NeuroImage 2004

Some examples



© Warner Bros. Adv. Media Services
Extraction des vecteurs de déplacement du standard H264 SVC



- Sports analysis
- Autonomous driving
- ..

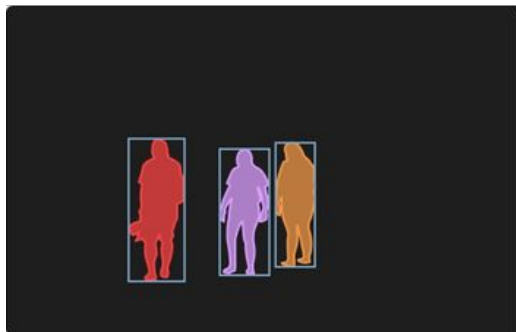
Three Flavors of Segmentation



Image



Semantic Segmentation

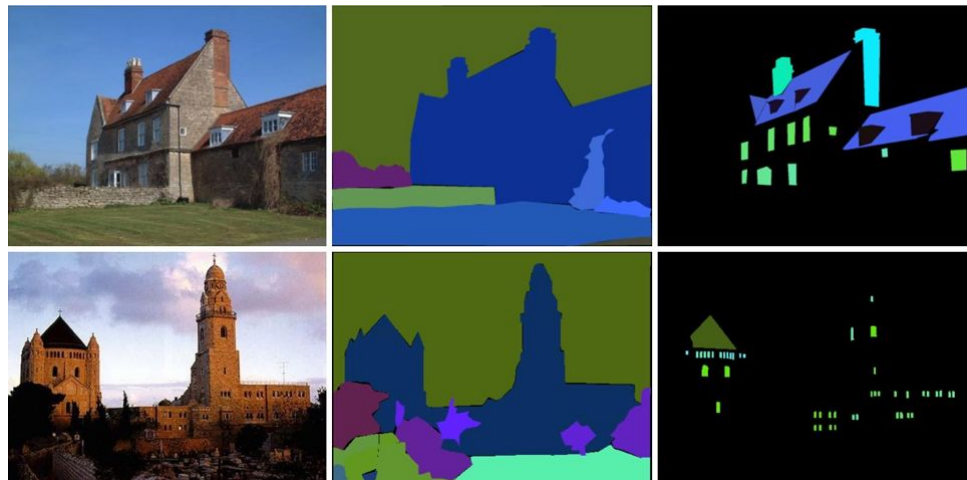


Instance Segmentation



Panoptic Segmentation

- **20K scene-centric images**, annotated with pixel-level objects and object parts labels.
- **150 semantic categories**: sky, road, grass, and discrete objects like person, car, bed



[Source:<https://www.kaggle.com/datasets/awsaf49/ade20k-dataset>]

Microsoft COCO: Common Objects in Context

Tsung-Yi Lin Michael Maire Serge Belongie Lubomir Bourdev Ross Girshick
James Hays Pietro Perona Deva Ramanan C. Lawrence Zitnick Piotr Dollár

Abstract—We present a new dataset with the goal of advancing the state-of-the-art in object recognition by placing the question of object recognition in the context of the broader question of scene understanding. This is achieved by gathering images of complex everyday scenes containing common objects in their natural context. Objects are labeled using per-instance segmentations to aid in precise object localization. Our dataset contains photos of 91 objects types that would be easily recognizable by a 4 year old. With a total of 2.5 million labeled instances in 328k images, the creation of our dataset drew upon extensive crowd worker involvement via novel user interfaces for category detection, instance spotting and instance segmentation. We present a detailed statistical analysis of the dataset in comparison to PASCAL, ImageNet, and SUN. Finally, we provide baseline performance analysis for bounding box and segmentation detection results using a Deformable Parts Model.

v:1405.0312v3 [cs.CV] 21 Feb 2015

1 INTRODUCTION

One of the primary goals of computer vision is the understanding of visual scenes. Scene understanding involves numerous tasks including recognizing what objects are present, localizing the objects in 2D and 3D, determining the objects' and scene's attributes, characterizing relationships between objects and providing a semantic description of the scene. The current object classification and detection datasets [1], [2], [3], [4] help us explore the first challenges related to scene understanding. For instance the ImageNet dataset [1], which contains an unprecedented number of images, has recently enabled breakthroughs in both object classification and detection research [5], [6], [7]. The community has also created datasets containing object attributes [8], scene attributes [9], keypoints [10], and 3D scene information [11]. This leads us to the obvious question: what datasets will best continue our advance towards our ultimate goal of scene understanding?

We introduce a new large-scale dataset that addresses

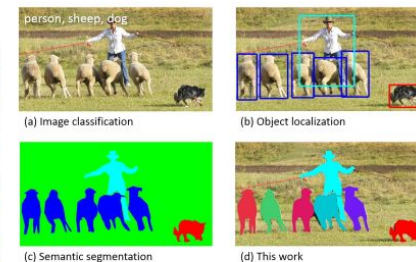


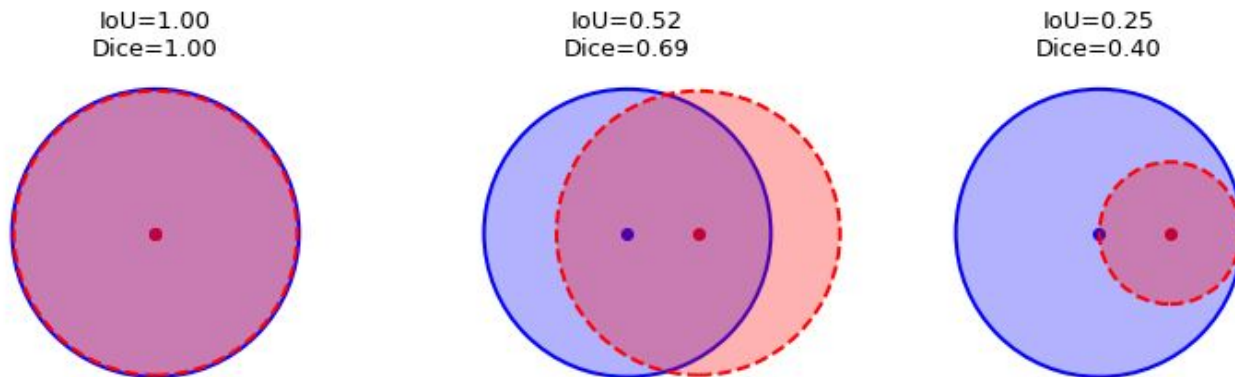
Fig. 1: While previous object recognition datasets have focused on (a) image classification, (b) object bounding box localization or (c) semantic pixel-level segmentation, we focus on (d) segmenting individual object instances. We introduce a large, richly-annotated dataset comprised of images depicting complex everyday scenes of common objects in their natural context.

[Source:<https://arxiv.org/abs/1405.0312>]

328K images

2.5M instances

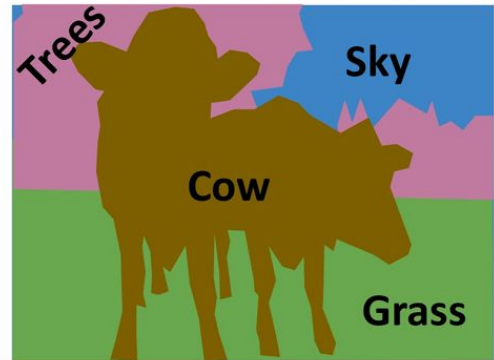
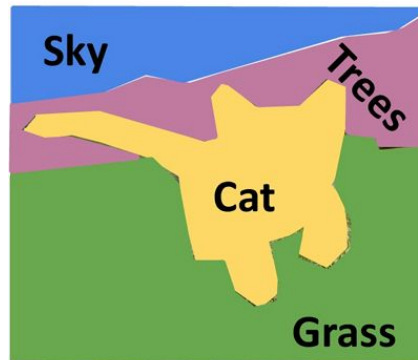
80 object category



[Source:<https://ilmontoux.github.io/2019/05/10/segmentation-metrics.html>]

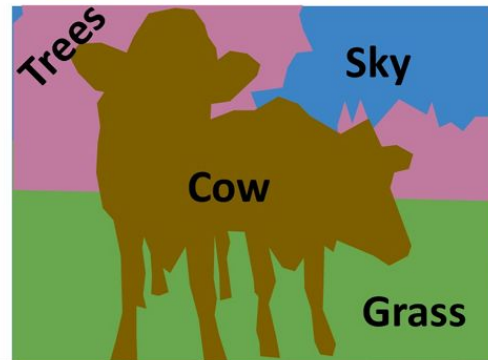
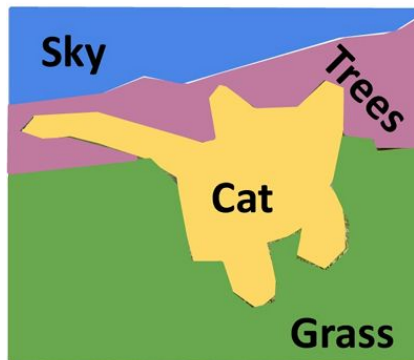
$$Dice(A, B) = \frac{2\|A \cap B\|}{\|A\| + \|B\|}, \quad Jaccard(A, B) = \frac{\|A \cap B\|}{\|A \cup B\|}$$

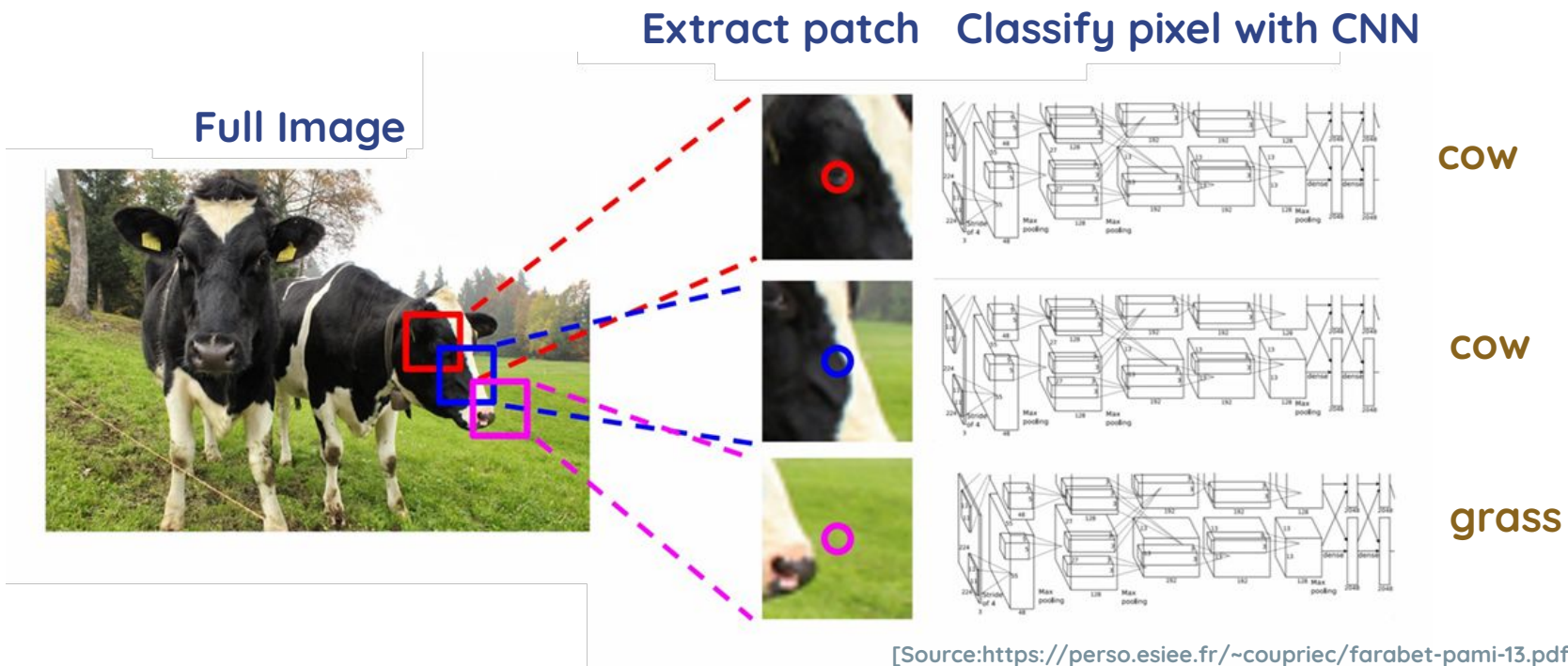
Label **each pixel** in the image
with a category label

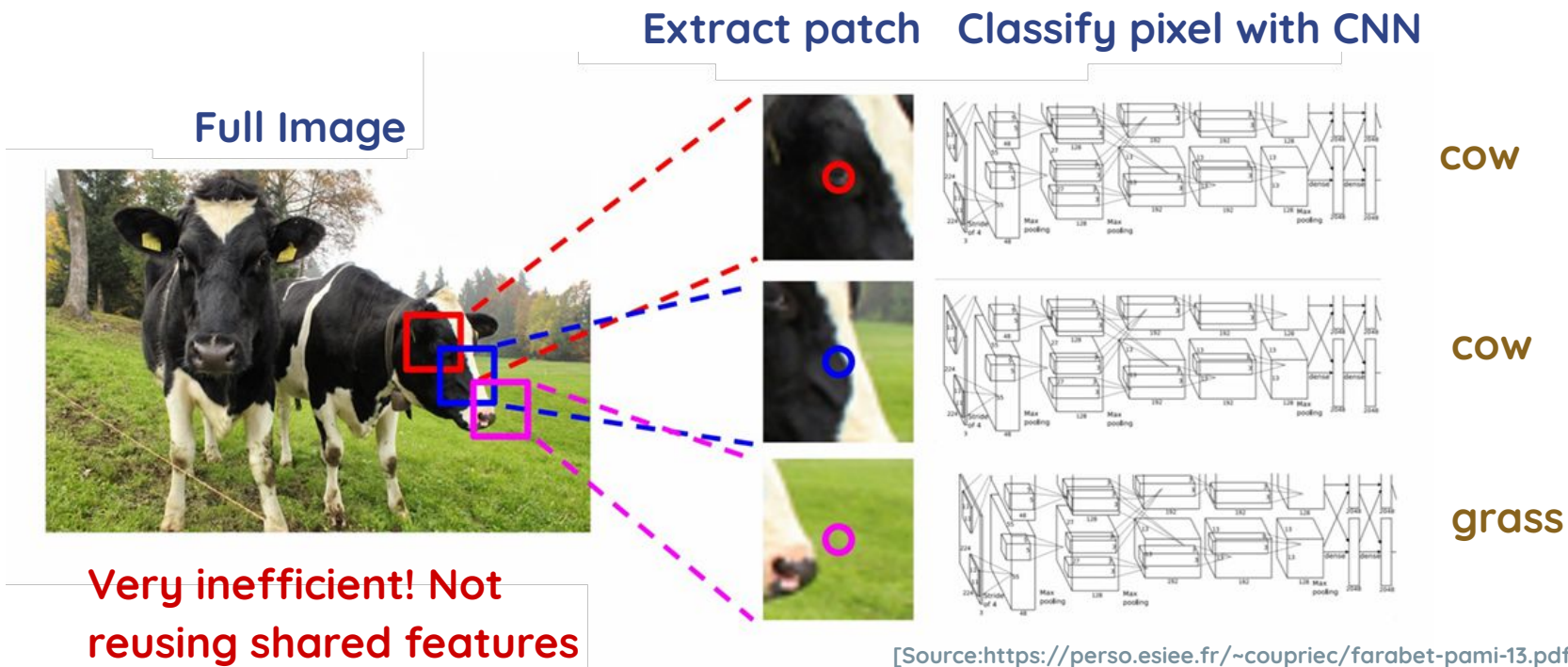


Label **each pixel** in the image
with a category label

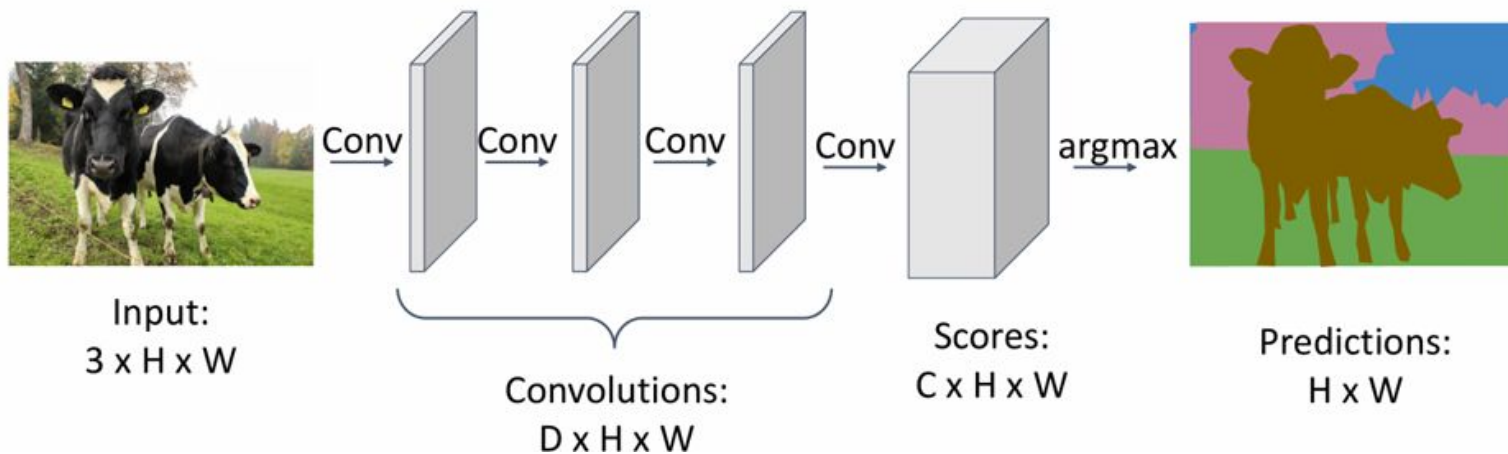
Don't differentiate **instances**,
only care about **pixels**







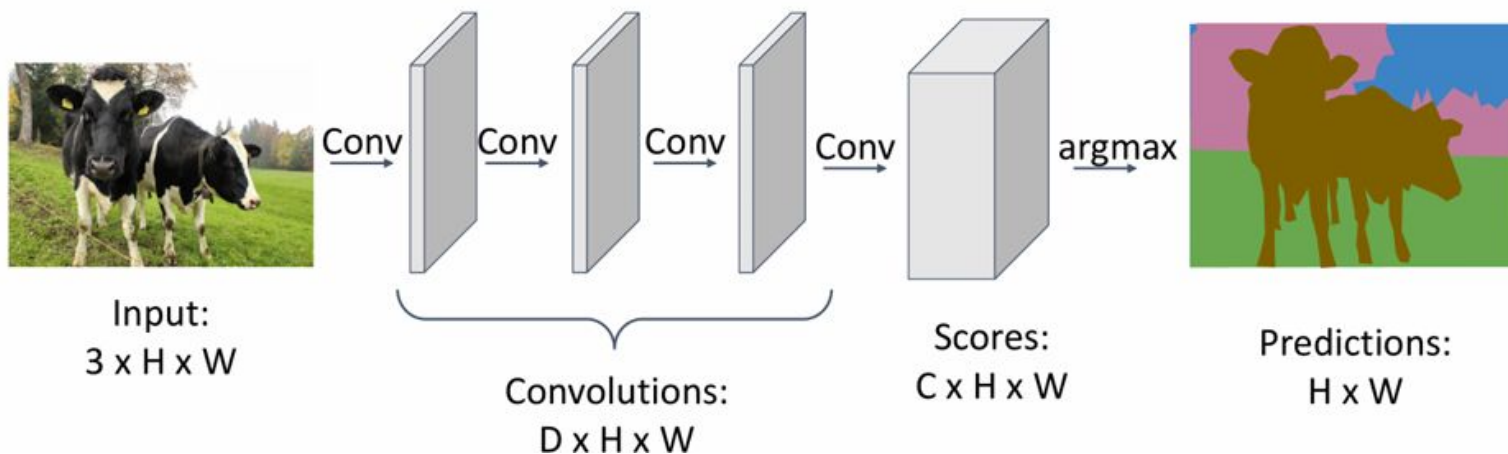
Design a network as a bunch of conv layers to make predictions for pixels at once



Loss Function : Per-Pixel cross-entropy

[Source: Long et al. 'Fully Convolutional Networks for Semantic Segmentation'
CVPR 2015]

Design a network as a bunch of conv layers to make predictions for pixels at once

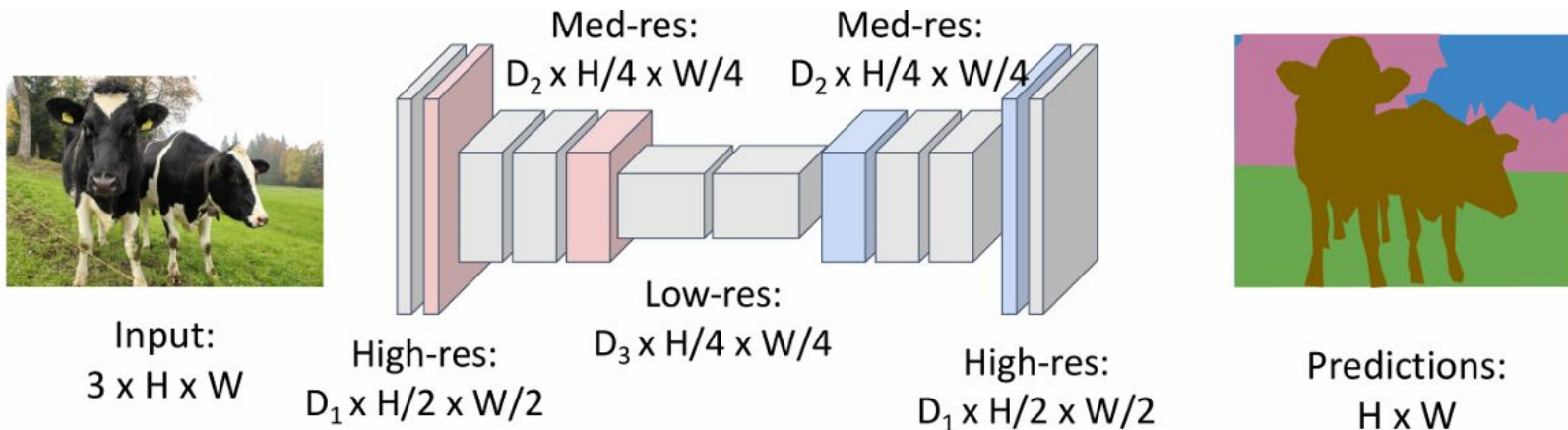


Problem: Convolution on high resolution images is expensive!

Loss Function : Per-Pixel cross-entropy

[Source: Long et al. 'Fully Convolutional Networks for Semantic Segmentation' CVPR 2015]

Design a network as a bunch of conv layers, with **downsampling** and **upsampling** inside the network

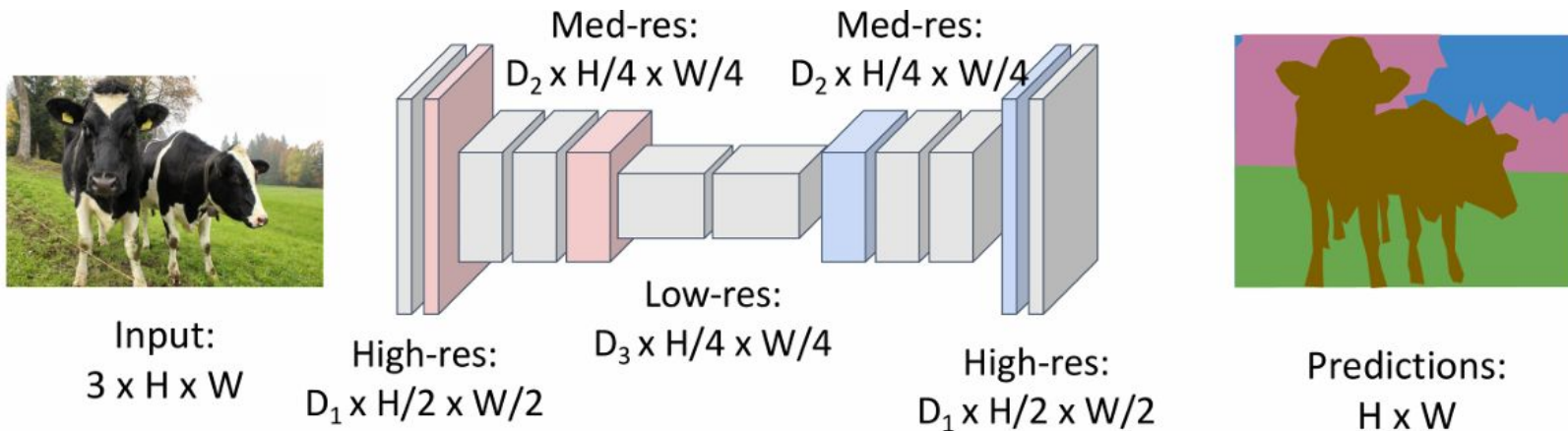


[Sources:

- Long et al. 'Fully Convolutional Networks for Semantic Segmentation' CVPR 2015
- Noh et al. Learning Deconvolution Network for Semantic Segmentation, ICCV 2015]

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Design a network as a bunch of conv layers, with **downsampling** and **upsampling** inside the network



Upsampling: ??

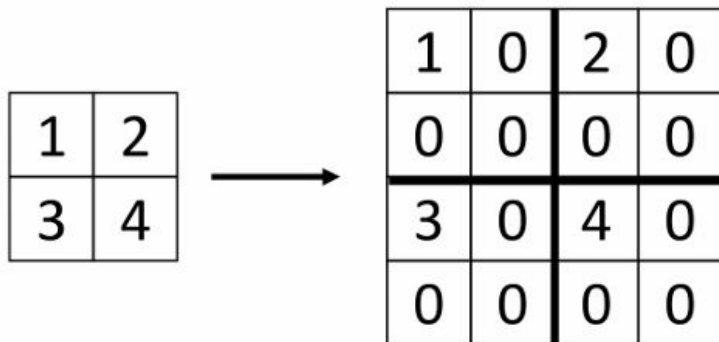
Downsampling: Pooling,
strided convolution

[Sources:

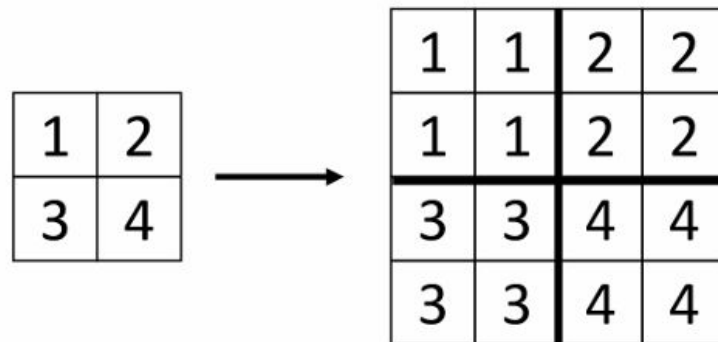
- Long et al. 'Fully Convolutional Networks for Semantic Segmentation' CVPR 2015
- Noh et al. Learning Deconvolution Network for Semantic Segmentation, ICCV 2015]

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Bed of Nails

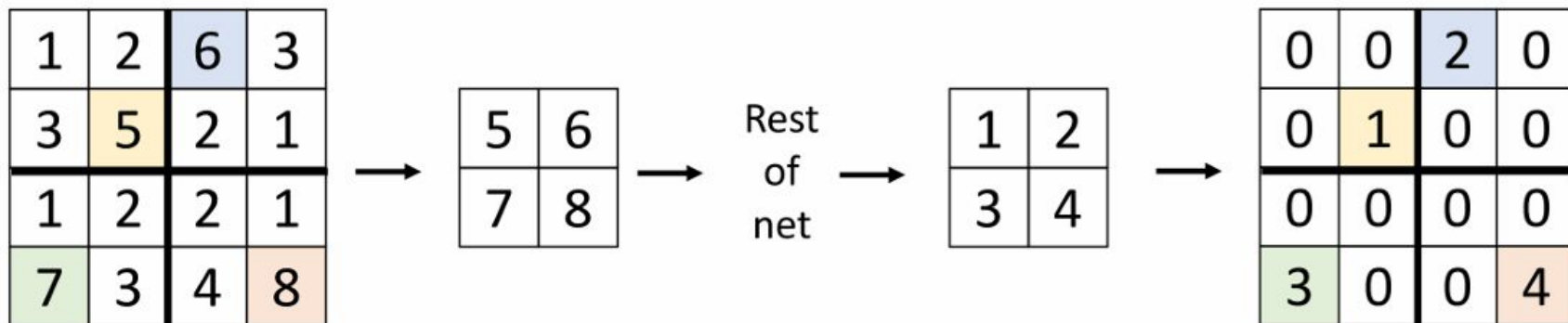


Nearest Neighbor

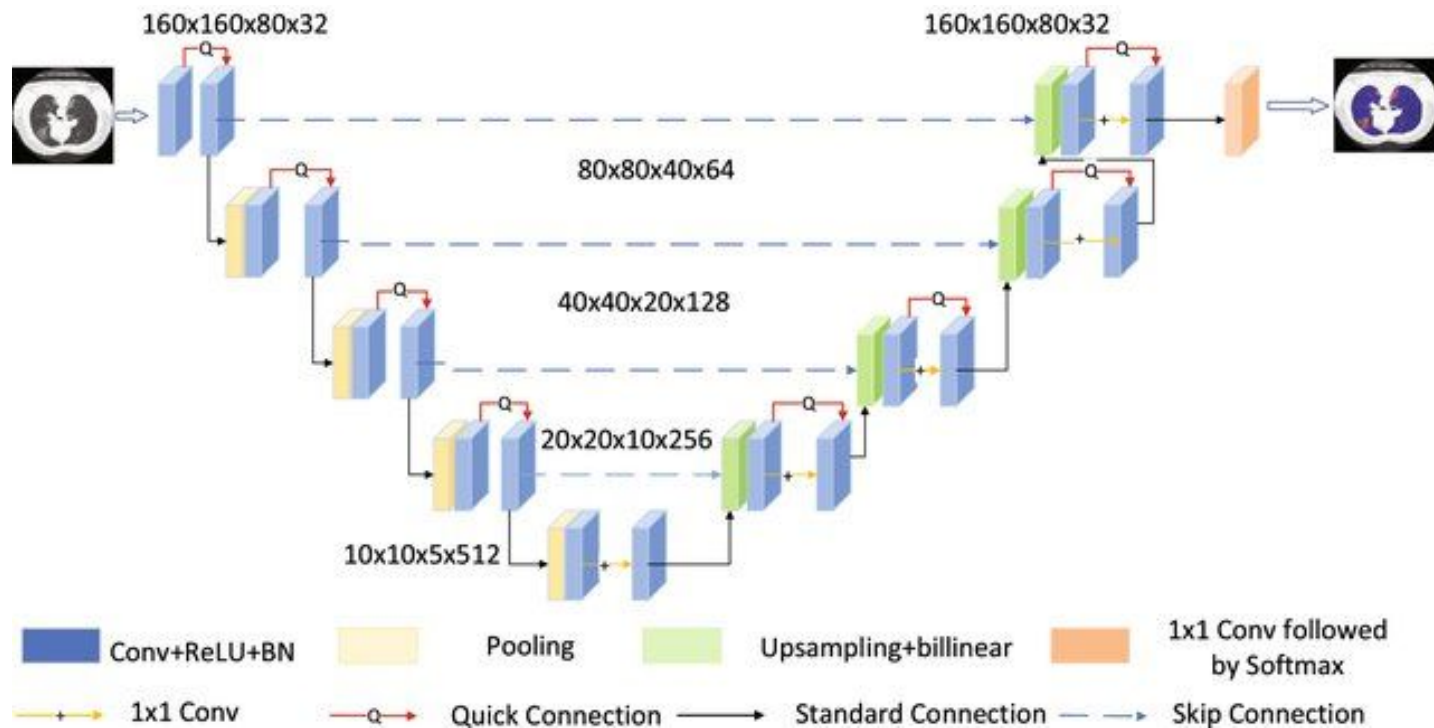


But in Network they used : **Max Unpooling**

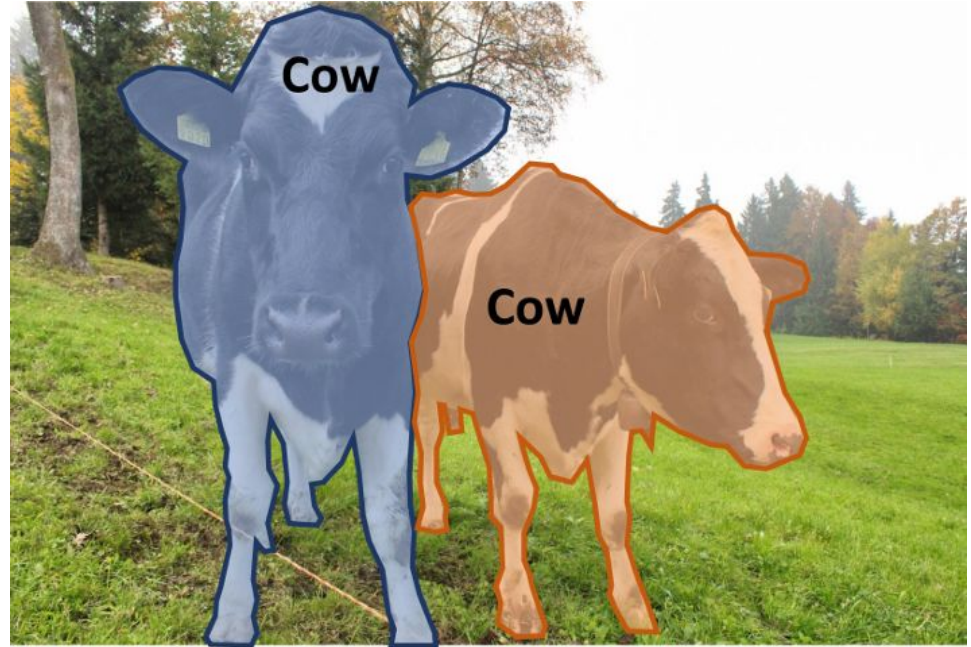
During the max pooling of the downsampling, remember which position had the max



During the max unpooling of the upsampling, place into remembered positions

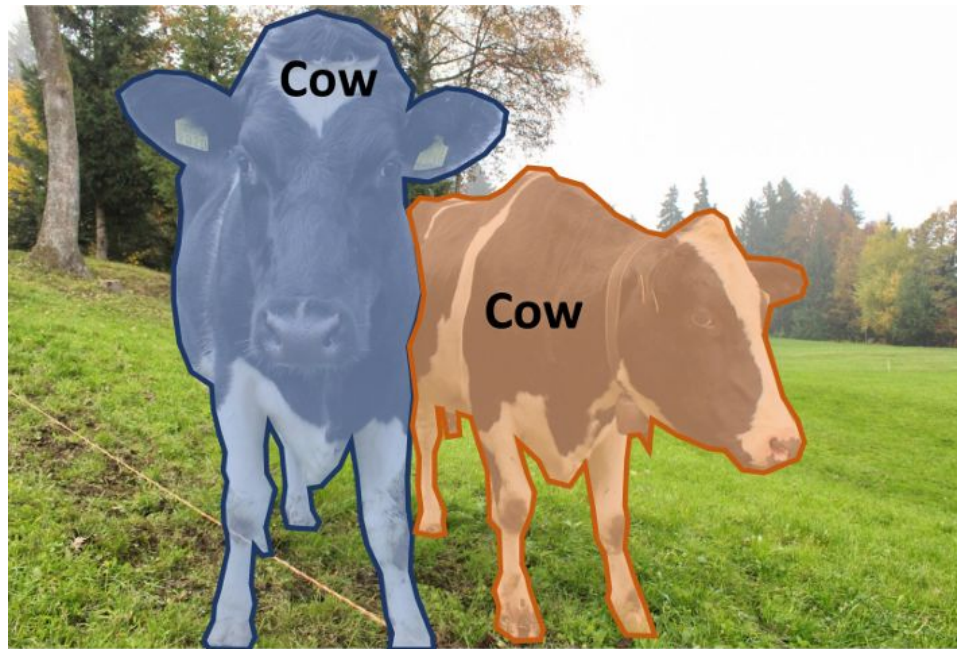


Detect all objects and
identify the **pixels** that
belong to each object



Detect all objects and
identify the **pixels** that
belong to each object

Approach : Object detection
then predict a segmentation
mask for each object



Mask R-CNN

Kaiming He Georgia Gkioxari Piotr Dollár Ross Girshick

Facebook AI Research (FAIR)

Abstract

We present a conceptually simple, flexible, and general framework for object instance segmentation. Our approach efficiently detects objects in an image while simultaneously generating a high-quality segmentation mask for each instance. The method, called Mask R-CNN, extends Faster R-CNN by adding a branch for predicting an object mask in parallel with the existing branch for bounding box recognition. Mask R-CNN is simple to train and adds only a small overhead to Faster R-CNN, running at 5 fps. Moreover,

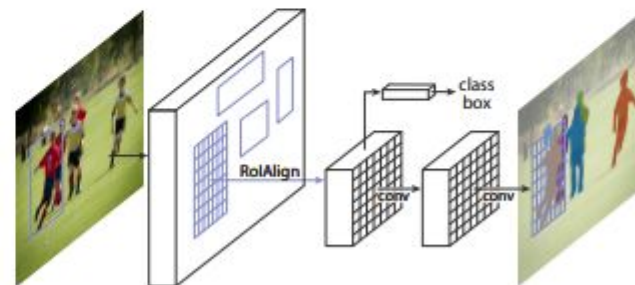
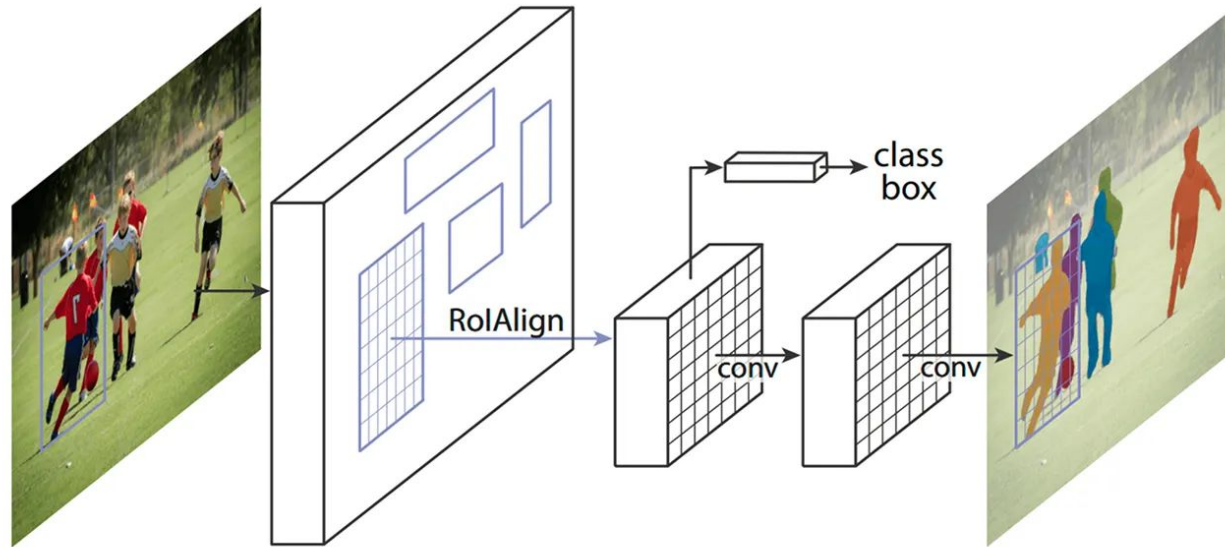


Figure 1. The **Mask R-CNN** framework for instance segmentation.

[Source: <https://arxiv.org/pdf/1703.06870> 2018]

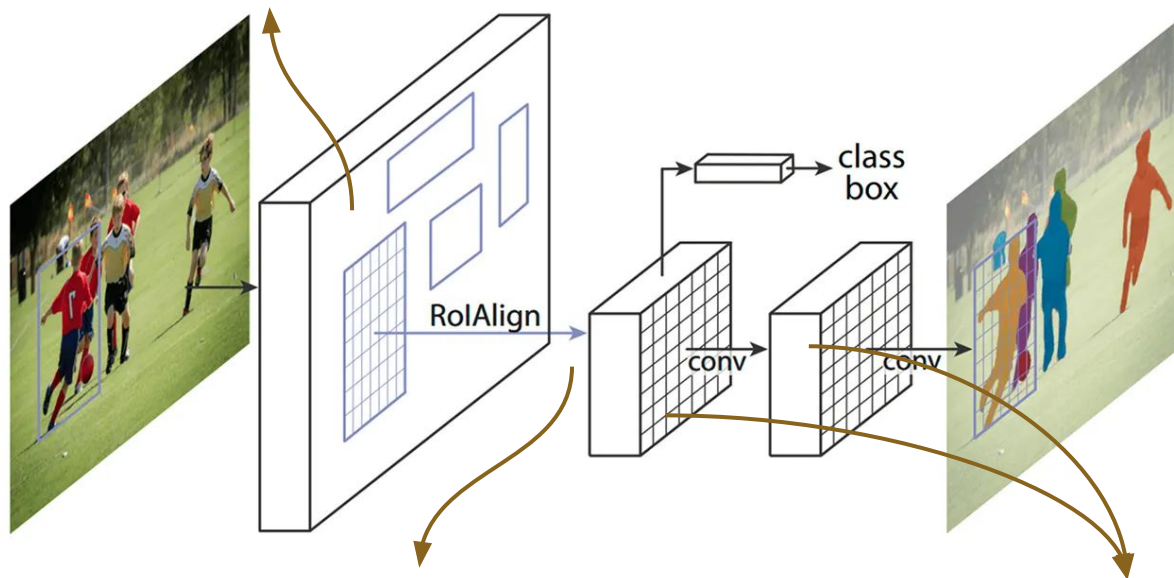
24 Jan 2018



[Source: <https://arxiv.org/pdf/1703.06870> 2018]

The backbone extracts features

[Source: <https://arxiv.org/pdf/1703.06870> 2018]

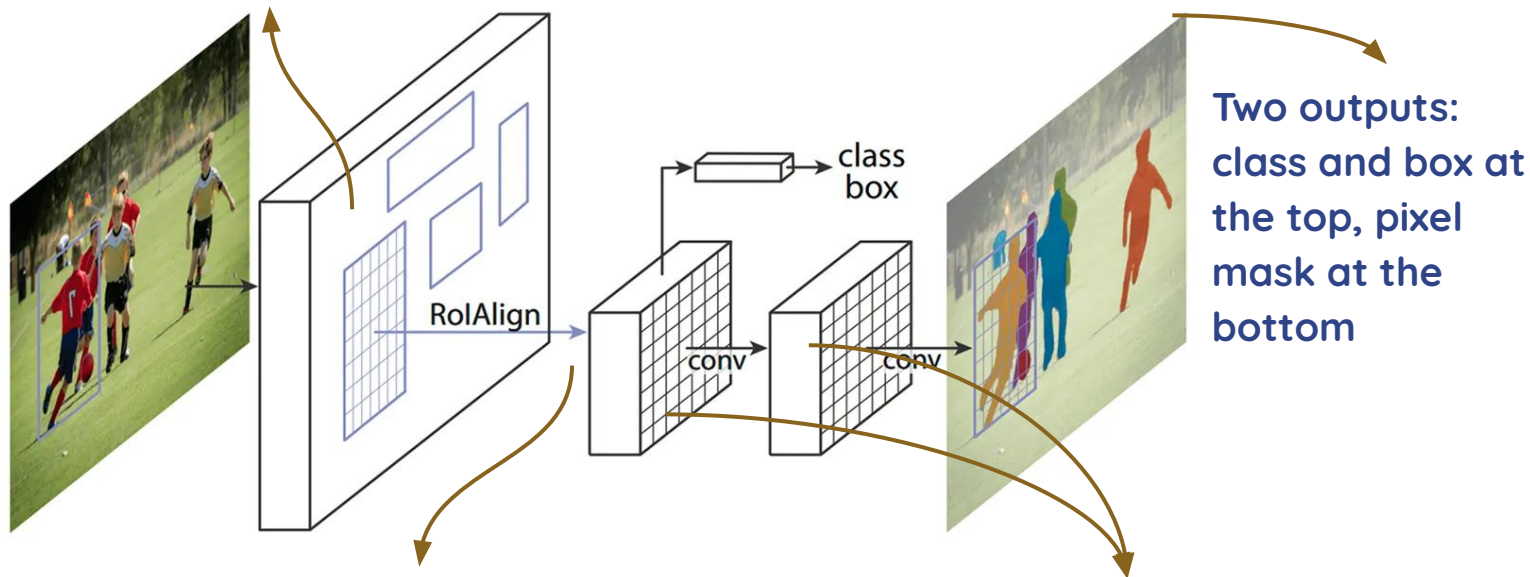


Region of Interest Align : crops and aligns the feature region for each detected object precisely

Two conv layers process each region.

[Source: <https://arxiv.org/pdf/1703.06870> 2018]

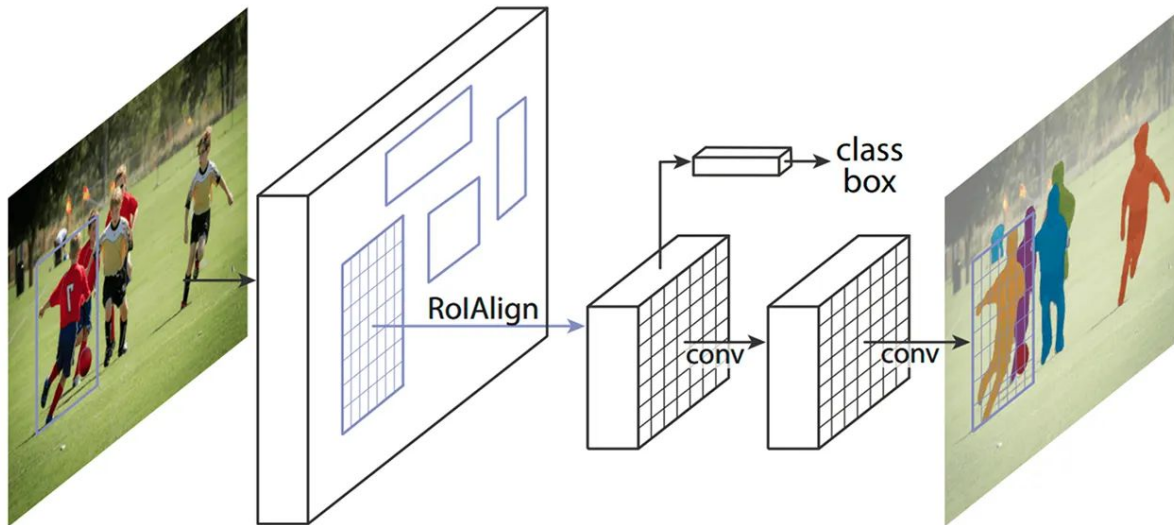
The backbone extracts features



Region of Interest Align : crops and aligns the feature region for each detected object precisely

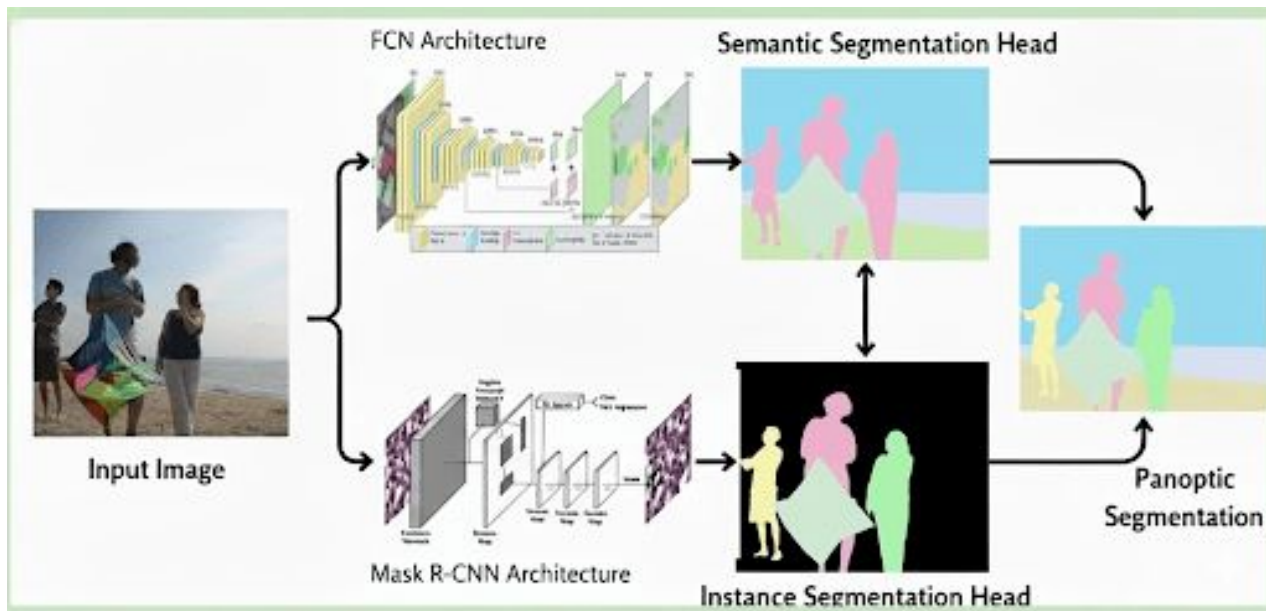
Two conv layers process each region.

[Source: <https://arxiv.org/pdf/1703.06870> 2018]



Problems :

- Still needs large annotated datasets, every mask drawn by hand
- Closed vocabulary, only detects the classes it was trained on, nothing else



Semantic; every pixel gets a class.
Instance; every object gets a mask.
Panoptic combines **both**.

[Source: <https://www.lightly.ai/blog/panoptic-segmentation>]

Every model we saw is a specialist. Change the domain, start over !

Every model we saw is a specialist. Change the domain, start over !

Segment Anything Model (SAM): the first foundation model for promptable segmentation.

Segment Anything

Alexander Kirillov^{1,2,4} Eric Mintun² Nikhila Ravi^{1,2} Hanzi Mao² Chloe Rolland³ Laura Gustafson³
Tete Xiao³ Spencer Whitehead Alexander C. Berg Wan-Yen Lo Piotr Dollár⁴ Ross Girshick⁴
¹project lead ²joint first author ³equal contribution ⁴directional lead
Meta AI Research, FAIR

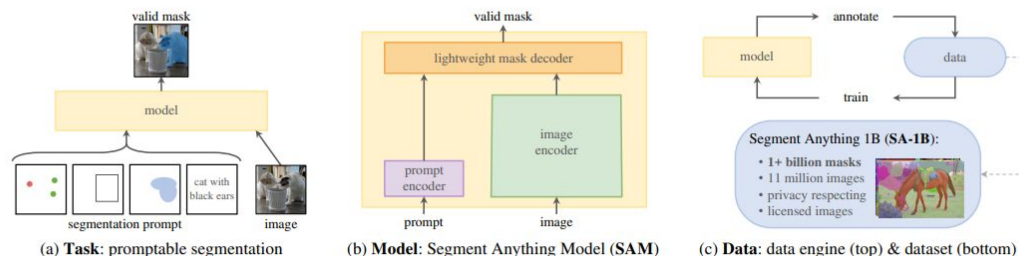
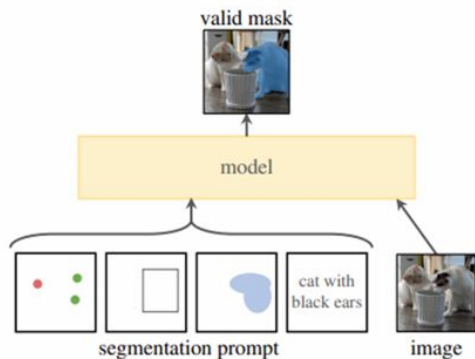


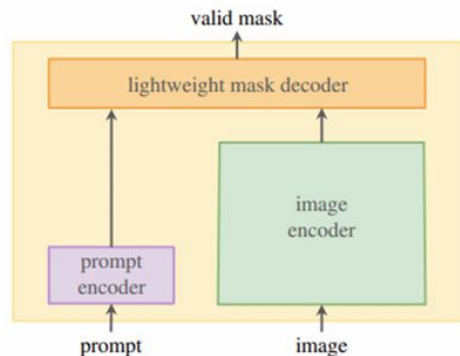
Figure 1: We aim to build a foundation model for segmentation by introducing three interconnected components: a promptable segmentation *task*, a segmentation *model* (SAM) that powers data annotation and enables zero-shot transfer to a range of tasks via prompt engineering, and a *data* engine for collecting SA-1B, our dataset of over 1 billion masks.

CVPR 2023 Apr 5

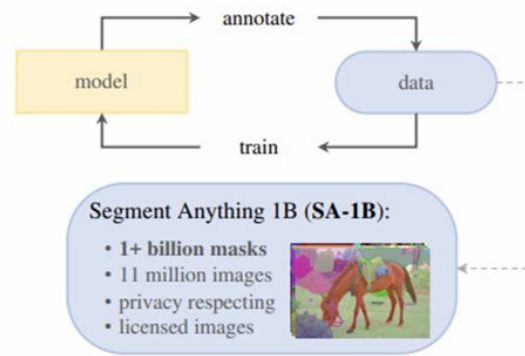
SAM is built with three interconnected components: A task, an model, and a data engine.



(a) **Task:** promptable segmentation



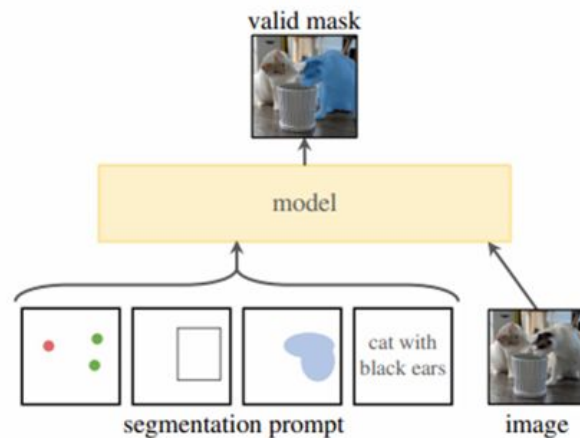
(b) **Model:** Segment Anything Model (SAM)



(c) **Data:** data engine (top) & dataset (bottom)

[Slide Credits: Qin Liu]

Given an **image** and a **prompt**, return a valid **mask**



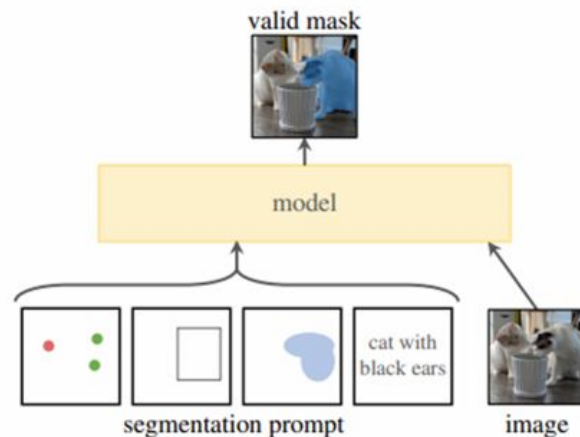
(a) **Task:** promptable segmentation

[Slide Credits: Qin Liu]

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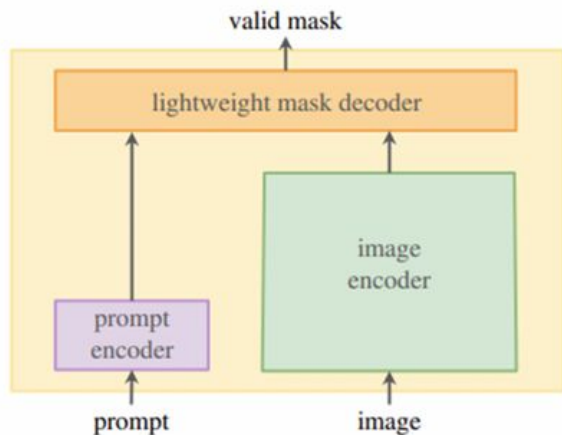
SAM accepts **two** kinds of prompts :

- **Sparse**: a click, a box, a text description. Minimal information, maximum flexibility
- **Dense**: a rough mask as prior. The user already has an approximation and wants SAM to refine it



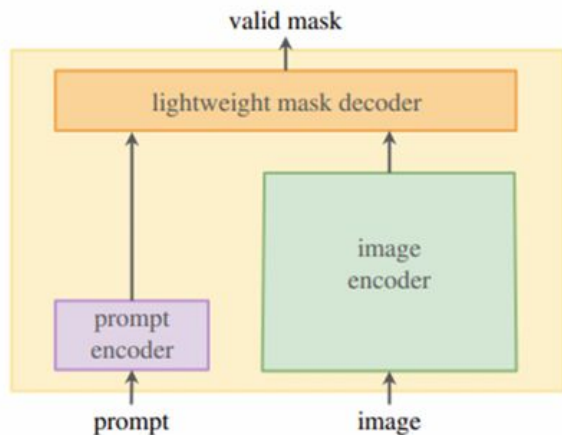
(a) **Task**: promptable segmentation

[Slide Credits: Qin Liu]

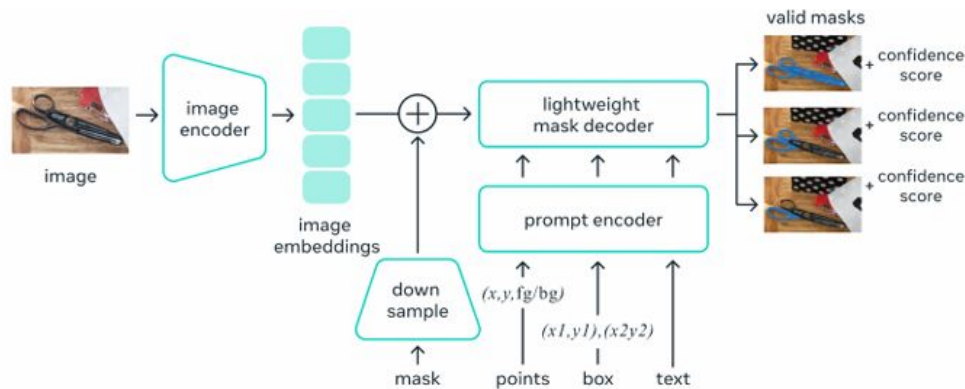


(b) **Model:** Segment Anything Model (SAM)

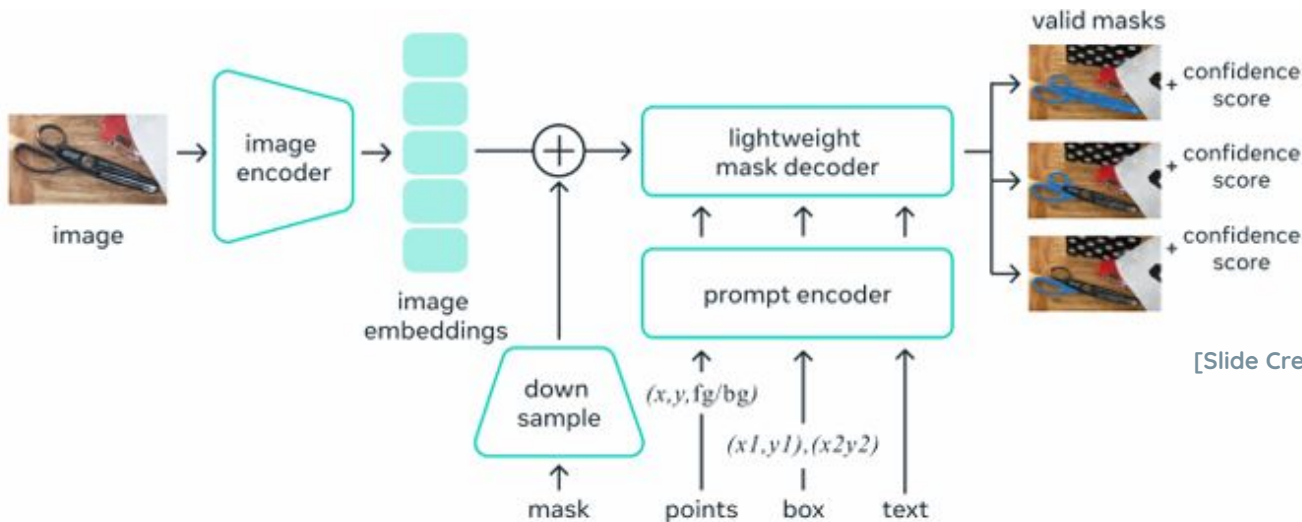
[Slide Credits: Qin Liu]



(b) **Model:** Segment Anything Model (SAM)

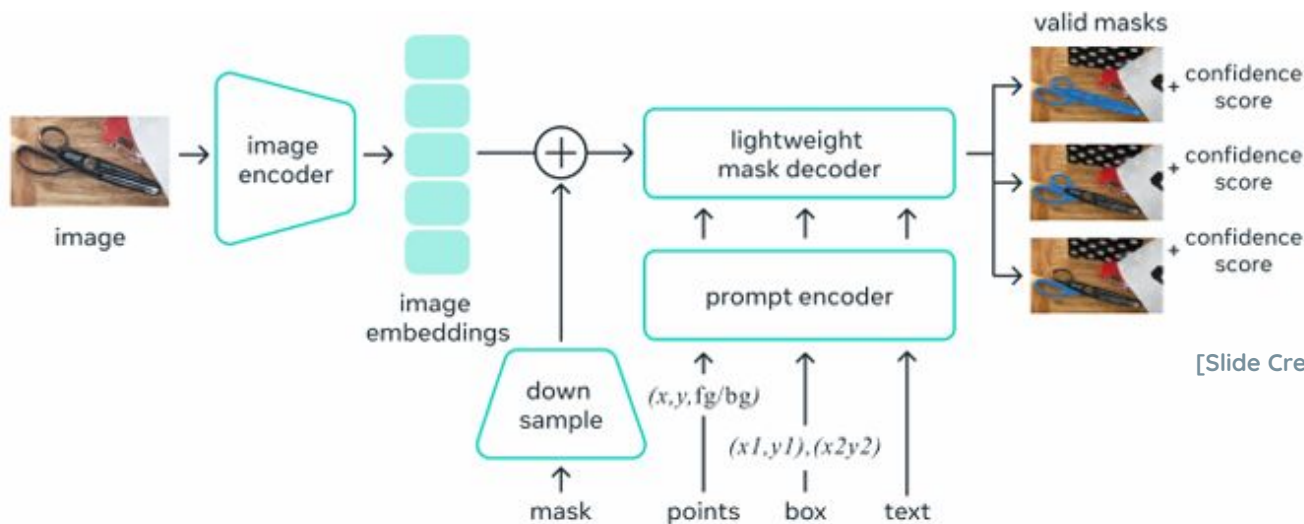


[Slide Credits: Qin Liu]



[Slide Credits: Qian Liu]

- A heavyweight **image encoder** outputs an image embedding.
- A lightweight **prompt encoder** converts the prompt into an embedding
- A **mask decoder** fuses image and prompt embeddings to produce masks



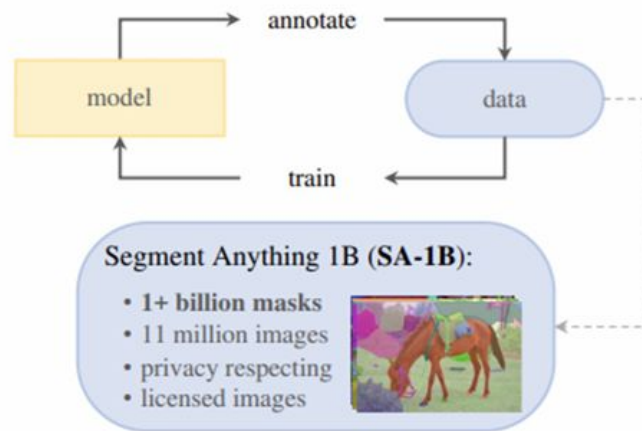
[Slide Credits: Qin Liu]

- A heavyweight **image encoder** outputs an image embedding.
- A lightweight **prompt encoder** efficiently queries the image embedding.
- A lightweight **mask decoder** produces object masks and confidence scores

You cannot train a foundation model on existing segmentation datasets.

COCO has 1.5 million masks.

ADE20k has a few hundred thousand



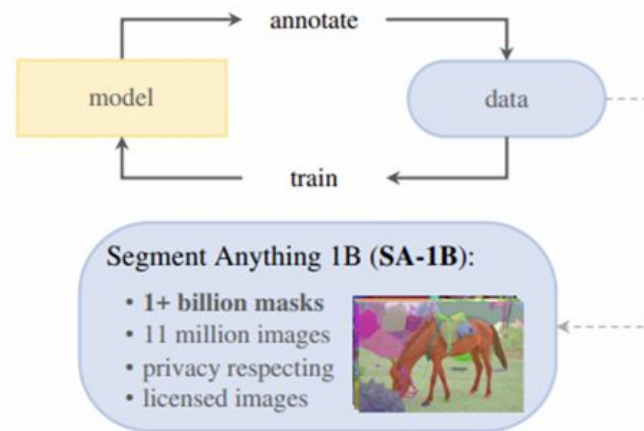
(c) **Data:** data engine (top) & dataset (bottom)

You cannot train a foundation model on existing segmentation datasets.

COCO has ~860K masks.

ADE20k has a few hundred thousand

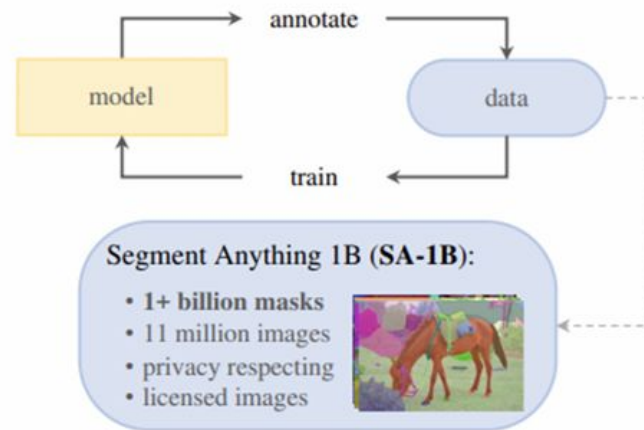
SAM needed something much **larger**. So Meta built the data engine, a system where SAM **itself generates** its own training data.



(c) **Data:** data engine (top) & dataset (bottom)

Three phases :

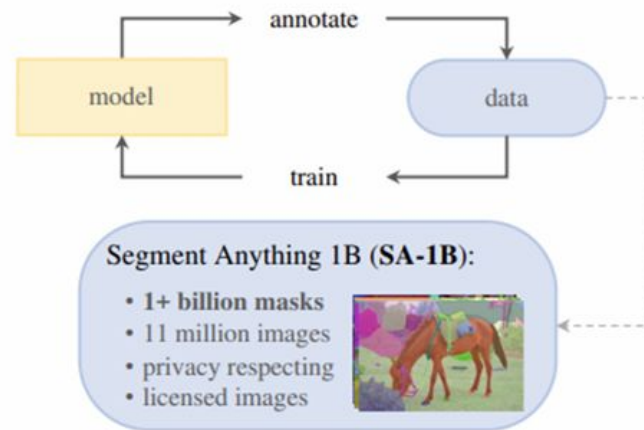
- **Phase 1:** Human assisted Annotators use SAM to label faster. They correct, SAM proposes.



(c) **Data:** data engine (top) & dataset (bottom)

Three phases :

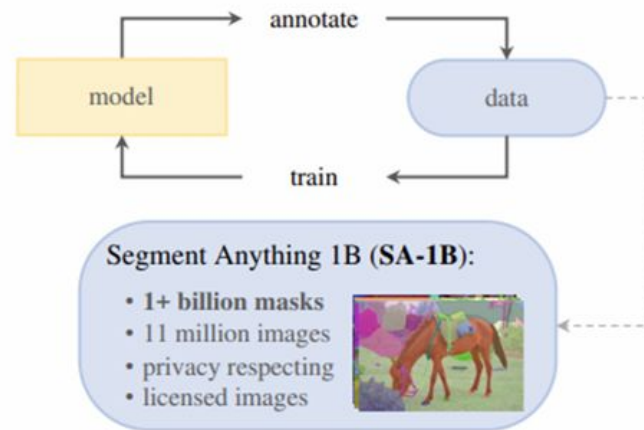
- **Phase 1:** Human assisted Annotators use SAM to label faster. They correct, SAM proposes.
- **Phase 2:** Semi-automatic SAM proposes confident masks. Humans only fill the gaps.



(c) **Data:** data engine (top) & dataset (bottom)

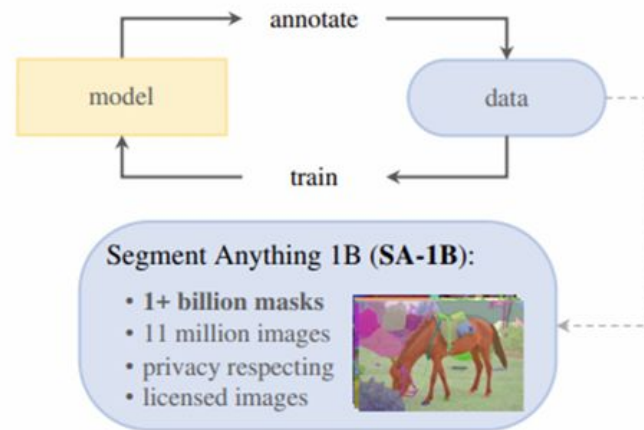
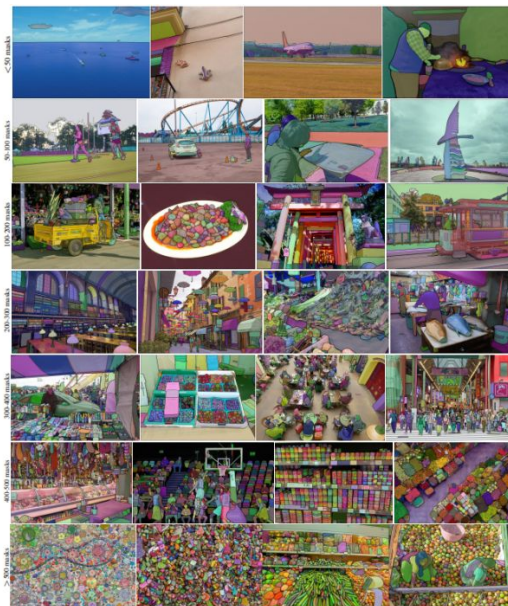
Three phases :

- **Phase 1:** Human assisted Annotators use SAM to label faster. They correct, SAM proposes.
- **Phase 2:** Semi-automatic SAM proposes confident masks. Humans only fill the gaps.
- **Phase 3:** Fully automatic SAM runs alone. No human at all.

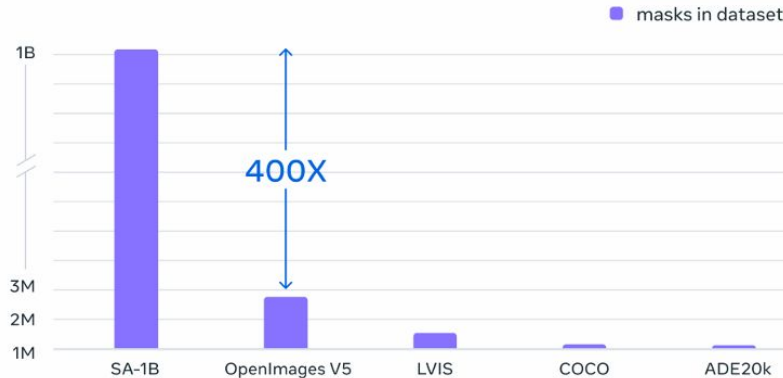
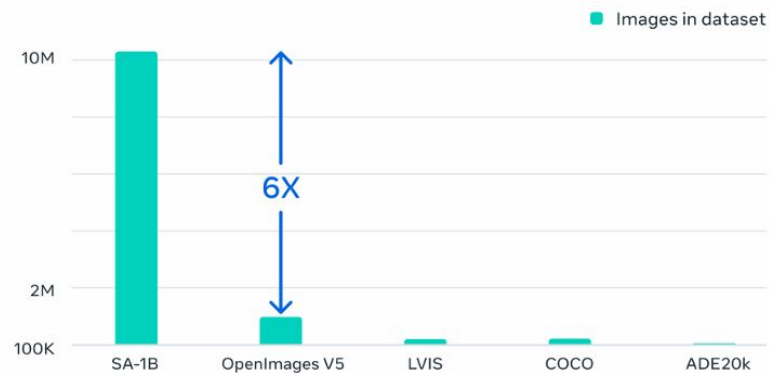


(c) **Data:** data engine (top) & dataset (bottom)

1.1 billion masks



(c) **Data:** data engine (top) & dataset (bottom)



400x more masks than any prior segmentation dataset

SAM (2023)

- First foundation model for promptable segmentation
- Image encoder ViT-H, prompt encoder, mask decoder
- SA-1B: 11M images, 1.1B masks
- Images only

SAM (2023)

- First foundation model for promptable segmentation
- Image encoder ViT-H, prompt encoder, mask decoder
- SA-1B: 11M images, 1.1B masks
- Images only

SAM2 (2024)

- Extends SAM to video
- Adds a memory bank, stores information from previous frames
- Faster

SAM (2023)

- First foundation model for promptable segmentation
- Image encoder ViT-H, prompt encoder, mask decoder
- SA-1B: 11M images, 1.1B masks
- Images only

SAM2.1 (2024)

- Improved model checkpoints
- Better performance on complex scenes
- Same architecture, better training

SAM (2023)

- First foundation model for promptable segmentation
- Image encoder ViT-H, prompt encoder, mask decoder
- SA-1B: 11M images, 1.1B masks
- Images only

SAM3 (2025)

- Prompt with an example image → finds all similar objects

SAM (2023)

- First foundation model for promptable segmentation
- Image encoder ViT-H, prompt encoder, mask decoder
- SA-1B: 11M images, 1.1B masks
- Images only

SAM3.1 (March 2026)

- Optimized for real-time video processing
- Faster, more accessible



Figure 1 SAM 3 improves over SAM 2 on promptable *visual* segmentation with clicks (left) and introduces the new promptable *concept* segmentation capability (right). Users can segment all instances of a visual concept specified by a short noun phrase, image exemplars (positive or negative), or a combination of both.



Books

- Computer Vision: Models, Learning, and Inference (2012). <https://amzn.to/2rxrdOF>
- Programming Computer Vision with Python (2012). <https://amzn.to/2QKTaAL>
- Multiple View Geometry in Computer Vision (2004). <https://amzn.to/2LfHLE8>
- Computer Vision: Algorithms and Applications (2010). <https://amzn.to/2Lclt4J>
- Computer Vision: A Modern Approach (2002). <https://amzn.to/2rv7AqJ>
- Deep Learning for Computer Vision : Image Classification, Object Detection and Face Recognition in Python(2019). <https://machinelearningmastery.com/deep-learning-for-computer-vision/>
- Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow — Aurélien Géron (2019)
<https://amzn.to/3uZbN6Z>
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